

CLINICAL AI MODEL VALIDATION POLICY

Project: Diabetes Readmission Clinical AI

Project Descriptor: 30-Day Hospital Readmission Clinical Decision-Support Prototype

Document Type: Clinical AI Model Validation Policy

Document Owner: AI Governance / Clinical AI Validation Function

Policy Status: Approved Project Governance Standard

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Classification: Project Governance Document

Review Cycle: Annual or upon material model change

Production Status: Not Authorized for Production Clinical Use

1. PURPOSE

This policy establishes the mandatory validation requirements for artificial intelligence and machine-learning models developed, evaluated, or considered for clinical decision-support use within the **Diabetes Readmission Clinical AI** project.

The policy ensures that model performance, safety, reliability, generalizability, fairness, explainability, and clinical suitability are evaluated through a controlled and evidence-based process before any progression toward clinical implementation.

Validation shall determine whether a model performs sufficiently and consistently for its **defined intended use and validation stage**. Completion of technical validation shall not, by itself, constitute authorization for production deployment or establish clinical effectiveness.

2. SCOPE

This policy applies to predictive models, preprocessing pipelines, feature-engineering components, operating thresholds, model updates, recalibrated models, and materially changed versions developed or evaluated under the Diabetes Readmission Clinical AI project.

It applies across the following validation stages:

1. Internal validation;
2. Locked-test validation;
3. Temporal validation;
4. External or geographic validation;
5. Prospective clinical validation; and
6. Revalidation following material change.

The policy applies to all personnel responsible for model development, validation, clinical review, governance, deployment assessment, and monitoring.

3. VALIDATION PRINCIPLES

Clinical AI validation shall comply with the following principles.

3.1 Intended-Use Alignment

Every validation shall evaluate the model against its formally defined clinical purpose, target population, prediction outcome, prediction horizon, intended user, and intended workflow.

Performance outside the validated intended use shall not be assumed.

3.2 Independence of Validation

Validation data shall be appropriately separated from model development activities.

The model shall not be optimized, retrained, recalibrated, or have its operating threshold changed using locked validation or external validation data before the prespecified evaluation has been completed.

3.3 Frozen Evaluation Configuration

Confirmatory validation shall use a documented and frozen configuration including, as applicable:

- model version;
- preprocessing pipeline;
- feature definitions;
- input schema;
- model parameters;
- operating threshold;
- outcome definition; and
- evaluation methodology.

Any material modification shall be subject to change control and may require revalidation.

3.4 Evidence Proportionality

Validation depth shall be proportional to the model's clinical significance, potential patient impact, degree of automation, uncertainty, and intended deployment environment.

3.5 Human Oversight

Validation of a clinical decision-support model shall not transfer clinical decision authority to the model.

Healthcare professionals remain responsible for clinical assessment and decisions unless a separately governed and authorized use explicitly establishes otherwise.

3.6 Transparency

Validation reports shall document both favorable and unfavorable findings. Material limitations, performance deterioration, subgroup heterogeneity, uncertainty, and unresolved risks shall not be omitted.

4. ROLES AND ACCOUNTABILITY

Role	Accountability
Model Development Lead	Provides the frozen model, preprocessing pipeline, feature specification, development evidence, and technical documentation
Validation Lead	Defines and executes the validation protocol independently from model optimization and documents validation findings.
Clinical Lead	Reviews clinical relevance, intended use, outcome definition, safety implications, workflow implications, and clinical limitations.

AI Governance Lead	Confirms evidence completeness, traceability, risk disposition, change control, and governance-gate requirements
Data Governance / Data Steward	Confirms dataset provenance, authorization, quality, representativeness, lineage, and permitted use.
Model Owner	Maintains accountability for the model throughout validation, deployment consideration, monitoring, and retirement
Approval Authority	Determines whether evidence supports progression, conditional progression, remediation, or rejection

Where organizational resources permit, confirmatory and external validation should include reviewers who were not responsible for model development.

5. VALIDATION CLASSIFICATION

5.1 Internal Validation

Internal validation evaluates model performance using data originating from the development source population while maintaining appropriate separation between model development and evaluation.

Cross-validation, bootstrap validation, development-validation splits, and other approved methods may support internal validation.

5.2 Locked-Test Validation

Locked-test validation is a confirmatory internal evaluation performed against a dataset withheld from model development and model-selection activities.

Before access to locked-test outcomes, the model configuration and evaluation plan shall be frozen.

Locked-test data shall not be used retrospectively to optimize the evaluated model while retaining the same validation claim.

5.3 Temporal Validation

Temporal validation evaluates a frozen model using patients or encounters originating from a later time period than the development cohort.

Temporal validation provides evidence of performance over time but shall not automatically be represented as geographic external validation.

5.4 External Validation

External validation evaluates the frozen model using an appropriately independent population, institution, healthcare system, geography, or data source not used to develop the model.

Before evaluation, the following shall be documented:

- external dataset provenance;
- population and setting;
- eligibility criteria;
- predictor availability and mapping;
- outcome compatibility;
- differences from the development population;
- missing-data handling;
- data harmonization procedures; and

- prespecified analysis plan.

A material incompatibility between the external data and the frozen model's input or outcome contract shall be documented rather than concealed through uncontrolled transformation.

5.5 Prospective Clinical Validation

Prospective validation evaluates the model using data generated after model freeze under a prespecified protocol.

Where the model influences or is visible within clinical workflow, the evaluation shall additionally address human factors, workflow interaction, safety events, model overrides, unintended consequences, and clinical utility as appropriate.

Technical performance alone shall not be represented as evidence of clinical effectiveness.

6. MINIMUM VALIDATION REQUIREMENTS

Every confirmatory validation shall address the following domains where applicable.

6.1 Dataset and Population Integrity

The validation report shall document:

- data source and provenance;
- validation period;
- inclusion and exclusion criteria;
- sample size;
- outcome prevalence;
- missingness;
- relevant population characteristics;
- data-quality limitations; and
- relationship between development and validation populations.

For external validation, independence from the development dataset shall be explicitly assessed.

6.2 Predictor and Outcome Integrity

Required model inputs shall be evaluated for availability, definition, coding consistency, measurement differences, missingness, and temporal availability.

The validation outcome shall be clinically and temporally consistent with the intended prediction target.

Variables unavailable at the intended prediction time shall not be introduced to improve validation performance.

6.3 Discrimination

Appropriate discrimination measures shall be reported.

For binary clinical prediction models, these may include:

- ROC-AUC; and
- precision-recall AUC where class imbalance makes it clinically informative.

Discrimination metrics shall not be interpreted independently of calibration and operating-point performance.

6.4 Calibration

Where predicted probabilities are clinically relevant, validation shall assess agreement between predicted and observed risk using appropriate quantitative and graphical methods.

Material miscalibration shall be documented as a validation finding.

6.5 Operating-Point Performance

Where the model uses a decision threshold, performance at the frozen operating point shall be evaluated.

Relevant measures may include:

- sensitivity;
- specificity;
- positive predictive value;
- negative predictive value;
- F1 score;
- alert rate or workload;
- confusion matrix; and
- clinically relevant workload or efficiency measures.

A threshold shall not be changed after observing confirmatory validation results and still be represented as validation of the original frozen operating point.

6.6 Subgroup and Fairness Assessment

Performance shall be evaluated across clinically and operationally relevant subgroups where sample size and data quality permit.

Material heterogeneity shall be documented, investigated, and incorporated into the residual-risk assessment.

Absence of a statistically significant difference shall not automatically establish fairness.

6.7 Robustness

Validation shall assess model behavior under relevant perturbations, missingness patterns, data-quality conditions, population slices, workflow conditions, or other foreseeable variations.

Known failure modes shall be translated into safety controls, monitoring requirements, use restrictions, or remediation where appropriate.

6.8 Explainability

Where explainability is required for the intended use, validation shall assess whether explanation methods are technically appropriate, reproducible, and sufficiently interpretable for the intended users.

Model attribution shall not be represented as clinical causality.

6.9 Clinical Safety

Validation shall identify potential harms associated with:

- false-negative predictions;
- false-positive predictions;
- automation bias;
- inappropriate reliance;
- model use outside intended scope;
- subgroup performance differences; and
- known model failure modes.

Material safety concerns shall be entered into the project's risk and governance process.

7. VALIDATION PROTOCOL AND REPORT

A confirmatory validation shall operate under a documented protocol established before evaluation of the final validation outcomes.

The protocol shall define, at minimum:

- validation objective;
- model and version;
- intended use;
- dataset;
- population;
- outcome;
- predictors;
- frozen operating threshold, where applicable;
- primary and secondary performance measures;
- subgroup analyses;
- robustness analyses;
- statistical methods; and
- governance decision criteria.

Results shall be recorded in a controlled **Clinical AI Model Validation Report**.

The report shall distinguish observed evidence from interpretation and shall document limitations, residual risks, unresolved issues, and deviations from the validation protocol.

8. VALIDATION GOVERNANCE GATE

Validation shall result in one of the following governance dispositions:

Disposition	Meaning
PASS	Progression may occur subject to documented controls, restrictions, remediation, additional evidence, or monitoring requirements.
CONDITIONAL PASS	Progression may occur subject to documented controls, restrictions, remediation, additional evidence, or monitoring requirements
REMEDiation REQUIRED	Material deficiencies require corrective action and appropriate revalidation before progression.
FAIL	Evidence does not support progression for the proposed intended use.
NOT ASSESSABLE	Available evidence is insufficient to determine the relevant validation outcome.

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A **PASS** or **CONDITIONAL PASS** applies only to the validation stage evaluated.

Neither disposition independently constitutes authorization for production clinical deployment.

9. EXTERNAL VALIDATION AND MODEL ADAPTATION

External validation shall evaluate the frozen model before adaptation whenever scientifically and operationally feasible.

If external validation identifies material performance deterioration, the result shall be documented as observed. The original validation result shall not be replaced by performance obtained after retraining, recalibration, threshold modification, or feature redesign.

Subsequent adaptation may be undertaken under controlled model-development and change-management processes.

A materially adapted model shall receive:

- updated model identity/version where required;
- documented rationale for modification;
- renewed validation;
- updated risk assessment; and
- governance approval before further progression.

10. REVALIDATION AND CHANGE CONTROL

Revalidation shall be considered when any of the following occurs:

- model retraining;
- material recalibration;
- operating-threshold change;
- feature addition, removal, or material redefinition;
- preprocessing change;
- material change in target population;
- material workflow change;
- material data-source or coding change;
- significant performance or calibration drift;
- newly identified subgroup or safety risk;
- material software or infrastructure change affecting model behavior;
- significant clinical practice change; or
- governance or regulatory requirement.

The scope of revalidation shall be proportional to the nature and potential impact of the change.

Changes affecting the validated model configuration shall be traceable through formal change control.

11. DOCUMENTATION, TRACEABILITY AND EVIDENCE RETENTION

Validation evidence shall be reproducible and traceable to the evaluated model.

The validation record shall include, as applicable:

- model identifier and version;
- model and preprocessing artifact identity;
- validation protocol;
- dataset provenance;
- feature and outcome definitions;
- evaluation code/version;
- performance results;
- subgroup analyses;
- robustness evidence;
- explainability evidence;
- safety findings;
- deviations;
- residual risks;
- governance disposition; and
- reviewer/approval records.

Evidence shall be retained in accordance with applicable project, organizational, legal, regulatory, and data-governance requirements.

12. CURRENT PROJECT APPLICATION

For the **Diabetes Readmission Clinical AI — 30-Day Hospital Readmission Clinical Decision-Support Prototype**, the current validation state is:

Validation Element	Current Status
Model	XGBoost
Model ID	DIABETES_READMISSION_XGB_D13_V1
Model Version	1.0.0
Frozen Operating Threshold	0.12
Internal Locked-Test Validation	Completed
External Validation	Not Established
Clinical Effectiveness	Not Established
Production Clinical Deployment	Not Authorized

The completed locked-test evaluation shall not be represented as external validation.

Any future external validation shall use the frozen model configuration unless a formally documented validation protocol specifies otherwise. Model adaptation following external evaluation shall be treated as a separate controlled activity.

13. POLICY EXCEPTIONS

Exceptions to this policy shall be documented and approved before implementation.

An exception record shall identify:

- requirement affected;
- justification;
- duration;
- associated risk;
- compensating controls;
- accountable owner; and
- approving authority.

Policy exceptions shall not be used to conceal material validation deficiencies or to represent incomplete evidence as established clinical performance.

14. REVIEW AND APPROVAL

This policy shall be reviewed at least annually and whenever material changes occur to the model lifecycle, intended use, validation methodology, applicable governance requirements, or clinical deployment strategy.

Approval of this policy establishes the validation governance requirements for the project. It does **not** authorize the Diabetes Readmission Clinical AI model for production clinical use.

Approval Record

Role	Name	Signature	Date
Project / Model Owner			
Clinical Validation Lead			
AI Governance Lead			
Independent Reviewer			
Approval Authority			
